

Appendices

Working Memory Mapping for WM-Dialogue

Table 1: Working Memory (WM) Processes and their Operationalizations in ITS Dialogue

WM Process	Operational evidence in ITS traces	Dialogue realization in WM-Dialogue	Functional role in this study
Encoding	First attempt on a new problem/step; first exposure to a new sub-goal; onset of attention to key constraints.	Problem introduction + initial student attempt; optional student re-statement of goal/known facts.	Defines the starting WM state and initial retrieval of declarative knowledge.
Maintenance	Reuse of the same intermediate result/constraint across consecutive steps; stable attempt pattern.	Repeated student attempts that preserve intermediate values; teacher confirmation.	Supports judging state stability and continuity of a procedural trace.
Monitoring	Incorrectness event (correctness=0) and/or negative feedback; repeated failure; hint request.	Teacher feedback indicating mismatch; student revision prompt (uncertainty/clarification).	Marks cognitive turning points: from stable execution to error detection.
Updating	Next attempt changes (value/operator) and moves toward correctness; incorporation of hint content.	Teacher hint/scaffold followed by student revised attempt; explicit “use-hint” turn.	Captures feedback-driven WM update and integration of new information.
Inhibition	Previously repeated erroneous rule disappears; switch away from recurring error type.	Teacher explicitly discourages a wrong rule; student stops using the prior pattern.	Prevents perseveration (buggy rule), reflecting inhibitory control.
Reconfiguration	Structural change in action sequence: step order changes, new method adopted, or new sub-goal.	Student produces a new plan; teacher provides higher-level guidance (not local correction).	Translates WM updates into actionable procedural trajectories.
Load / Stuck	Multiple consecutive incorrect attempts (≥ 2), high hint frequency, lack of progress.	Student expresses being stuck; teacher escalates scaffolding (step decomposition).	Operationalizes capacity limits and sustained difficulty; aligns with cognitive load patterns.
External Support	Hint/feedback events; step decomposition prompts; system-provided reminders of constraints.	Turns labeled as teacher hint / scaffold; optionally include hint level (general $\rightarrow$ specific).	Treats ITS guidance as external WM support for pedagogical scaffolding.

Human Verification Rubric

Table 2: Human Verification Rubric for Data Quality

Validation Dimension	Core Indicators	Fail Conditions
Factual Accuracy	Mathematical logic, calculation results, problem constraints.	Calculation errors; misuse/misapplication of concepts or rules; ignoring key problem constraints leading to invalid reasoning.
Evidence Consistency	Consistency between the dialogue and the evidence trace.	Inconsistencies between the dialogue and the known interaction trace; hallucinating facts that are not present in the trace.
Cognitive Plausibility	Smoothness of state transitions; match between ability level and tier; explainability of the updating process.	“Epiphany” jumps (sudden correction without hints); tier mismatch (low-tier student using expert strategies); forced smoothness.
Pedagogical Appropriateness	Appropriate scaffolding intensity; reasonable guidance strategies; proper emotional support.	Teacher directly provides answers (non-scaffolding); misleading guidance; overly robotic tone (lacking empathetic/affective support).

Evaluation Questionnaire Design and Scoring Standards with Descriptions

Table 3: Evaluation Questionnaire Design and Scoring Standards with Descriptions

<i>Questionnaire Design</i>	
Q1: Does the model understand the key requirements of the question? <i>Evaluate whether the model shows a full comprehension of the task and problem requirements provided by the teacher.</i>	
Q2: Does the model effectively apply the teacher's feedback? <i>Evaluate whether the model makes appropriate use of the teacher's feedback during problem-solving and adjusts its approach accordingly, reflecting a proper understanding and application of the teacher's suggestions.</i>	
Q3: Does the model demonstrate real-time cognitive processing ability during problem solving? <i>Evaluate whether the model exhibits the ability to process information in real time, including how it handles new problems or feedback by quickly adjusting its thinking and selecting the right strategies.</i>	
Q4: Does the model effectively manage cognitive load, breaking down and processing information logically? <i>Evaluate whether the model manages cognitive load effectively, breaking down complex tasks into manageable steps, completing the problem step by step, and avoiding confusion or unreasonable information processing.</i>	
Q5: Does the model's response align with the student's cognitive level? <i>Evaluate whether the model's responses match the cognitive level it simulates, demonstrating an understanding and complexity consistent with the actual level of the student.</i>	
<i>Scoring Standards</i>	
0: Not Matching	0.75: Mostly Matching
0.25: Somewhat Not Matching	1: Completely Matching
0.5: Generally Matching	

Table 4: An example of the WM-Dialogue dataset.

Turn	Role	Content
1	Teacher	Let's start today's number line task. Your first task is to find the position of 0.6. Try moving along the number line and tell me your starting point.
2	Student	Alright, I'll start by looking at 0.5 since 0.6 is close to it. I think I should move slightly to the right from 0.5 to get to 0.6. Let's see, I moved to 0.606.
3	Teacher	That's very good! You've stopped at 0.606, which is within the range of the correct answer. Well done! Now, let's move on to the second challenge. Can you find the position of 0.03?
4	Student	Sure, I think 0.03 is quite close to 0.0. I'll start at 0 and move slightly to the right. I think... maybe a tiny bit further... Okay, I've stopped at 0.019.
5	Teacher	Excellent! You've found 0.019, which is very close to 0.03 and falls within our acceptable range. You're doing a great job. Now for the final challenge, can you determine which is larger: 0.08 or 0.018?
6	Student	Hmm, 0.08 seems larger since it's closer to 0.1 than 0.018 is. I'll choose 0.08 as the larger number.
7	Teacher	That's correct! You've done really well. You correctly identified 0.08 as the larger number. You've successfully completed all the challenges today. What did you learn from these tasks?
8	Student	I learned that estimating and comparing numbers on a number line can help find specific points more easily. Moving step by step and comparing them to nearby landmarks like 0.5 or 0.1 really helps.